Paper Study

Improved techniques for measurement of superconductivity in diamond anvil cells by magnetic susceptibility

Student: Hsieh, Chang-En
Advisors: Prof. Chen, Yang Yuan; Dr. Ou, Min-Nan
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Inductive modulation method

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Principles and Challenges

- High-frequency generator → excitation coil → alternating magnetic field → sample & signal coil → induced EMF in signal coil
- **superconducting state** → diamagnetism → reduced magnetic flux variation in signal coil → **decreased EMF**
- EMF change ≈ background EMF (same order of magnitude)

  → **magnetic flux through sample ≈ magnetic flux through signal coil (same order of magnitude)**

  → detectable superconducting transition signal (difficult to fulfill)

For example, the background electromotive force originating in the coil having 4 mm in diameter with 300 turns under action of the alternating magnetic field with frequency 10 kHz and amplitude 0.5 Oe will have a magnitude of about 1 V. However, the signal from the superconducting sample having a size about 100 μm does not exceed 100 nV.

Issues with the compensation coil and block

- The background reduction provided by the **compensating coil** is limited to a factor of **100–500**.

  → a slight **geometric asymmetry** in the positioning of the signal and compensating coils relative to the excitation coil.

  → **Eddy currents** generated in the metallic components distort the uniform distribution of magnetic flux.
- A compensating block with adjustable EMF amplitude and phase can reduce the background signal to the order of 10^{-6} V.

  → Temperature-induced changes in geometry and conductivity can make the background signal exceed the amplifier's dynamic range.

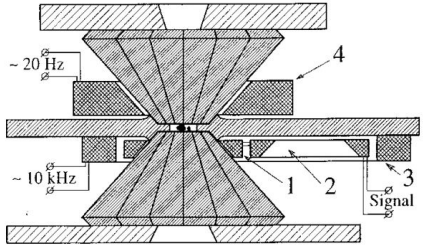
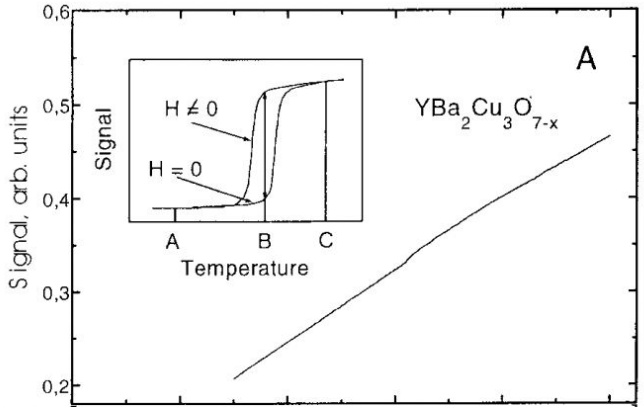


FIG. 1. Diamond anvils and coils system: signal coil (1), compensating coil (2), high frequency exciting coil (3), low frequency modulating coil (4). The sample is located inside the hole in the metallic gasket.

Modulation methods

- Near the transition temperature, a moderate magnetic field can exceed the critical value, shifting the superconducting transition to lower temperatures, as shown in the inset of Fig. A.
- An alternating magnetic field at frequency f suppresses superconductivity in both directions, producing amplitude modulation at 2f only near Tc, while not affecting background-related factors such as conductivity or geometry.

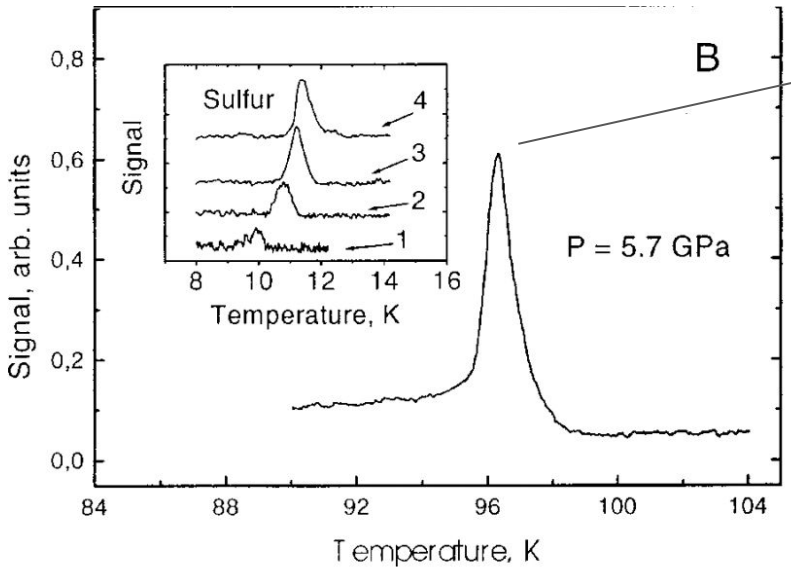


Measurement Process

1. **First Amplification** (Boosts the raw HF signal)
2. **Extraction of 2f Component** (Via LIA or Diode)
3. **Second Amplification** (Boosts the low-frequency modulation)
4. **Lock-in Detection** (Final filtering and signal recovery)
5. **Data Output** (Fed into computer for processing)

Measurement results

- low-frequency magnetic field increases → low-frequency signal increases
  → When the superconducting transition is shifted by more than the transition width, the superconducting transition becomes broadened.



The low-frequency signal causes a sharp change in magnetic susceptibility at Tc

Methods of increasing sensitivity

- Since the low-frequency signal is bounded by the change in the high-frequency amplitude during the superconducting transition, enhancing the EMF variation is essential for higher sensitivity.

When the amplitude of the high-frequency magnetic field exceeds the critical magnetic field at T = Tc - ΔT, the superconducting state is destroyed.

$$\Delta E = 4\pi f \mu_0 N n I_e d^3 / D_e D_s,$$

- Number of turns: 200–300
- Diameter: 9 to 11 mm

- Number of turns: 250–350 (optimal range)
- Diameter: Limited by the diamond anvil size
- More turns → higher impedance, harder amplifier matching, increased noise

Methods of increasing sensitivity

$$\Delta E = 4\pi f \mu_0 N n I_e d^3 / D_e D_s ,$$

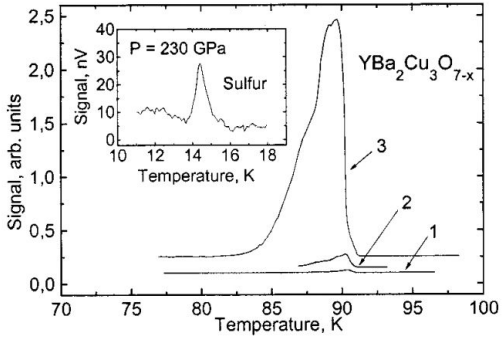


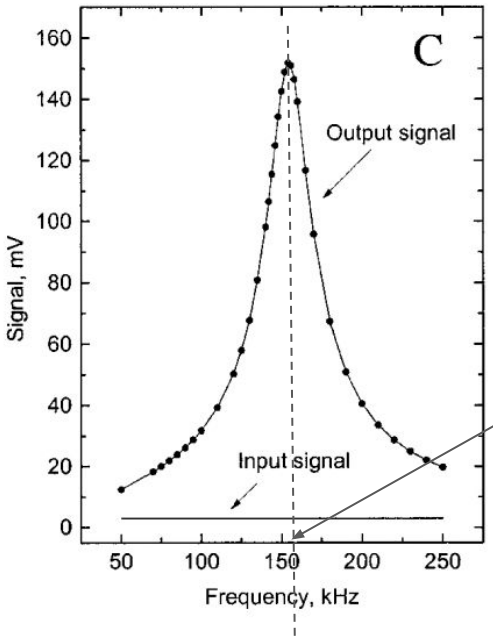
FIG. 5. Results of measurements at different frequencies on the standard sample of $YBa_2Cu_3O_{7-x}$ with linear dimension $d_s = 140\ \mu m$: (1) high frequency 10 kHz [signal/noise=21, $d_{min} = d_s$/(signal/noise)$^{1/3}$ = 50 μm], (2) high frequency 60 kHz (signal/noise= 110, d_{min}=3 μm), (3) resonant circuit with transformer operating at resonant frequency 145 kHz (signal/noise =2100, d_{min}=10 μm). Inset: Example of operating registering system with resonant circuit-superconductivity of sulfur at pressure 230 GPa.

- The signal-to-noise ratio increases approximately in proportion to the increase in frequency.
- The noise of the coil system may decrease when a smaller number of turns is used in the signal coil.

Resonance circuit

- High frequencies → increased output impedance → amplifier input impedance becomes insufficient.
- Impedance matching difficulties → limited transformation factor → Transformer output noise (from the coils) < amplifier input noise → loss of sensitivity.
- Solution to noise mismatch: a resonance circuit with a relatively small transformation factor; resonance provides output–input noise level matching. The voltage transmission factor of the circuit is 50.
- The total inductance of the signal and compensating coils causes only negligible changes in the resonance frequency during temperature variations in the high-pressure cell.

Resonance circuit



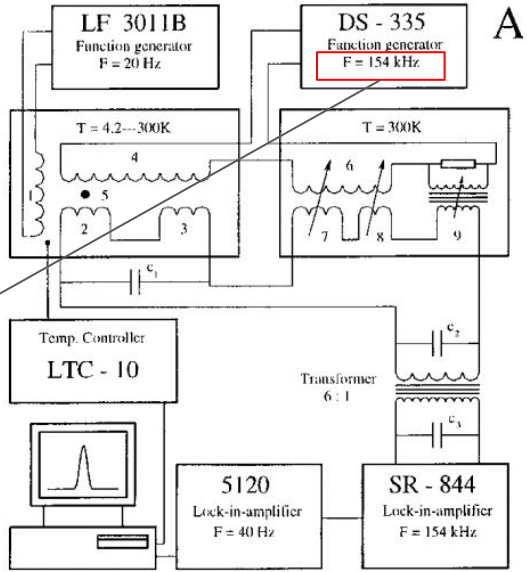


FIG. 3. (A) The block diagram of the experimental setup. Diamond anvil cell: low frequency modulating coil (1), signal coil (2), compensating coil (3), high frequency exciting coil (4), sample (5). Compensating block: additional high frequency excitation coil (6), additional compensating coils (7, 8), transformer (9) for compensation of the real component of the output signal. C_1—capacitance of the cable connecting system of coils with compensating block, C_2—capacitance of the cable connecting compensating block with transformer, C_3—input capacitance of the lock-in amplifier SR-844. (B) Equivalent scheme of resonant circuit. R—common resistance of